IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2024

MSc and EEE/EIE PART IV: MEng and ACGI

SYSTEMS IDENTIFICATION AND LEARNING

Monday, 13 May 10:00 am

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer FOUR questions.

All questions carry equal marks

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : F. Teng

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SYSTEMS IDENTIFICATION AND LEARNING

1. Consider the stochastic process $y(\cdot)$ described by the following recursive equation

$$y(t) = \frac{1}{4}w(t) + \frac{11}{30}w(t-1) + \frac{2}{15}w(t-2) - \frac{1}{4}y(t-1) + \frac{3}{8}y(t-2)$$

$$w(t) = -\frac{9}{2}e(t) - 3e(t-1) + 4e(t-2) - \frac{13}{10}w(t-1) - \frac{2}{5}w(t-2)$$

where $e(\cdot) \sim WN\left(1, \frac{2}{9}\right)$ is a non-zero mean white process.

a) Ascertain whether the process $y(\cdot)$ is stationary in steady state, and compute the expected value $m_y = \mathbb{E}[y(\cdot)]$ of the process $y(\cdot)$.

Hint for the student: *Write the process $y(\cdot)$ in terms of transfer functions in the $\mathscr{Z}$ transforms domain via an appropriate usage of the delay operator z^{-1}.*

[6 Marks]

b) Compute the auto-covariance function $\gamma_y(\tau)$ of the process $y(\cdot)$, for all $\tau \in \mathbb{Z}$.

Hint for the student: *Determine the spectral canonical form of the transfer functions determined in your answer to Question a).*

[7 Marks]

c) Determine the expression of the spectrum $\Gamma_y(\omega)$ of the process $y(\cdot)$, for all angular frequencies $\omega \in \mathbb{R}$.

[4 Marks]

d) Sketch the diagram of the spectrum $\Gamma_y(\omega)$.

[3 Marks]

2. Consider the stochastic process $y(\cdot)$ described by the following recursive equation:

$$y(t) = \frac{1}{2}y(t-1) - \frac{1}{8}y(t-2) + w(t-2) + \frac{2}{3}w(t-3) \tag{2.1}$$

where $w(\cdot) \sim WN(0,1)$ is a white process.

a) Compute the two samples $\gamma_y(0)$ and $\gamma_y(1)$ of the auto-covariance function $\gamma_y(\tau)$ of the stochastic process $y(\cdot)$.

[8 Marks]

b) Consider the following family of models:

$$\mathscr{M}_1\left(a,\lambda^2\right): \quad y(t) = ay(t-1) + e(t) \quad \text{with } e(\cdot) \sim WN(0,\lambda^2).$$

Assuming to have at your disposal an arbitrarily large number of measurements taken from the stochastic process $y(\cdot)$ described by the recursive equation (2.1), compute the parameters a° and λ^{2° of the family of models $\mathscr{M}_1$ that are optimal according the *prediction error minimisation* (PEM) criterion.

[6 Marks]

c) Consider the following family of models

$$\mathscr{M}_2\left(b,\mu^2\right): \quad y(t) = by(t-2) + \xi(t) \quad \text{with } \xi(\cdot) \sim WN(0,\mu^2).$$

Assuming to have at your disposal an arbitrarily large number of measurements taken from the stochastic process $y(\cdot)$ described by the recursive equation (2.1), compute the parameters b° and μ^{2° of the family of models $\mathscr{M}_2$ that are optimal according the *prediction error minimisation* (PEM) criterion.

[6 Marks]

3. Consider the stochastic processes $\alpha(\cdot)$ and $\beta(\cdot)$ described by the following two recursive equations, respectively:

$$\alpha(t+1) = \frac{3}{4}\alpha(t) + \frac{5}{6}\beta(t) + \frac{3}{4}e(t) + 2u(t)$$
$$\beta(t+1) = \frac{1}{3}\beta(t) + \frac{3}{2}e(t) + 3u(t)$$

where $e(\cdot) \sim WN\left(0, \frac{1}{4}\right)$ and the input $u(\cdot)$ is specified in each question formulated below.

a) Ascertain whether the processes $\alpha(\cdot)$ and $\beta(\cdot)$ are stationary in steady state **for each** of the following three definitions of the input $u(\cdot)$:

– Case 1: $u_1(\cdot) \sim WN(1,1)$ with $e(\cdot)$ and $u_1(\cdot)$ independent,
– Case 2: $u_2(t) = e^{-t}$, u_2 deterministic
– Case 3: $u_3(t) = \sin(t)$, u_3 deterministic.

Hint for the student: *Write the processes $\alpha(\cdot)$ and $\beta(\cdot)$ in terms of transfer functions in the $\mathscr{Z}$ transforms domain via an appropriate usage of the delay operator z^{-1}.*

[5 Marks]

b) Setting $u(t) = 0$, for all $t \in \mathbb{Z}$, write the recursive equation associated with the spectral canonical form of the process $\alpha(\cdot)$.

[4 Marks]

c) Setting $u(t) = 0$, for all $t \in \mathbb{Z}$, write the recursive equation of the two-steps ahead predictor $\widehat{\alpha}_0(t|t-2)$ of $\alpha(t)$ from measured past values of $\alpha(\cdot)$. In addition, compute the variance of the prediction error $\varepsilon_0(t) = \alpha(t) - \widehat{\alpha}_0(t|t-2)$.

[4 Marks]

d) With $u(\cdot)$ defined as a generic **deterministic** function, write the recursive equation of the two-steps ahead predictor $\widehat{\alpha}(t|t-2)$ of $\alpha(t)$ from measured past values of $\alpha(\cdot)$. In addition, compute the variance of the prediction error $\varepsilon(t) = \alpha(t) - \widehat{\alpha}(t|t-2)$.

[4 Marks]

e) Compare the variance of ε_0 obtained in your answer to Question c) with the variance of ε obtained in your answer to Question d) and discuss your findings.

[3 Marks]

4. The temperature T of a steel bar in a metal factory has to be measured but cannot be directly accessed by a temperature measurement sensor due to mechanical and thermal constraints.

Suppose that the temperature T can be related to other two physical quantities x and y via the following linear relationship:

$$T = x + y$$

Two sensors A and B are available and provide measurement a and b having the following characteristics, respectively:

$$\mathsf{A}: \quad a = \frac{1}{3}x + \frac{2}{3}y + e_a \qquad\qquad \mathsf{B}: \quad b = \frac{4}{3}x + \frac{1}{2}y + e_b$$

where the measurement errors e_a and e_b are **mutually independent random variables** with:

$$\mathbb{E}[e_a] = 0\,, \quad \mathrm{Var}[e_a] = \frac{1}{2}; \quad \mathbb{E}[e_b] = 1\,, \quad \mathrm{Var}[e_b] = \frac{1}{3}\,.$$

a) Consider a linear estimator of the form

$$\widehat{T} = \alpha a + \beta b + \gamma.$$

where α, β, and γ are constant parameters to be determined. Establish the conditions on the parameters α, β, and γ such that the estimator $\widehat{T}$ is unbiased.

[6 Marks]

b) Compute the values of α, β, γ such that $\widehat{T}$ is unbiased and the variance of the prediction error $\varepsilon = T - \widehat{T}$ is the smallest possible. Furthermore, compute the variance $\mathrm{Var}[\varepsilon]$ of the prediction error ε.

[7 Marks]

c) Suppose that $N > 1$ sensors of the type A are available each providing a measurement a_i, $i = 1,2,\ldots N$. Likewise, $M > 1$ sensors of the type B are also available each providing a measurement b_j, $j = 1,2,\ldots M$. Specifically:

$$\mathsf{A}:\ a_i = \frac{1}{3}x + \frac{2}{3}y + e_a^{(i)},\ i = 1,2,\ldots,N \quad \mathsf{B}:\ b_j = \frac{4}{3}x + \frac{1}{2}y + e_b^{(j)},\ j = 1,2,\ldots,M$$

where the measurement errors $e_a^{(i)}$ and $e_b^{(j)}$ are **mutually independent random variables** with:

$$\mathbb{E}\left[e_a^{(i)}\right] = 0\,, \quad \mathrm{Var}\left[e_a^{(i)}\right] = \frac{1}{2}; \quad \mathbb{E}\left[e_b^{(j)}\right] = 1\,, \quad \mathrm{Var}\left[e_b^{(j)}\right] = \frac{1}{3}\,.$$

Consider the estimator

$$\widetilde{T} = \widetilde{\alpha}\sum_{i=1}^{N} a_i + \widetilde{\beta}\sum_{j=1}^{M} b_j + \widetilde{\gamma}$$

where $\widetilde{\alpha}$, $\widetilde{\beta}$, and $\widetilde{\gamma}$ are constant parameters to be determined. Establish the conditions on the parameters $\widetilde{\alpha}$, $\widetilde{\beta}$, $\widetilde{\gamma}$ such that the estimator $\widetilde{T}$ is unbiased.

[7 Marks]